

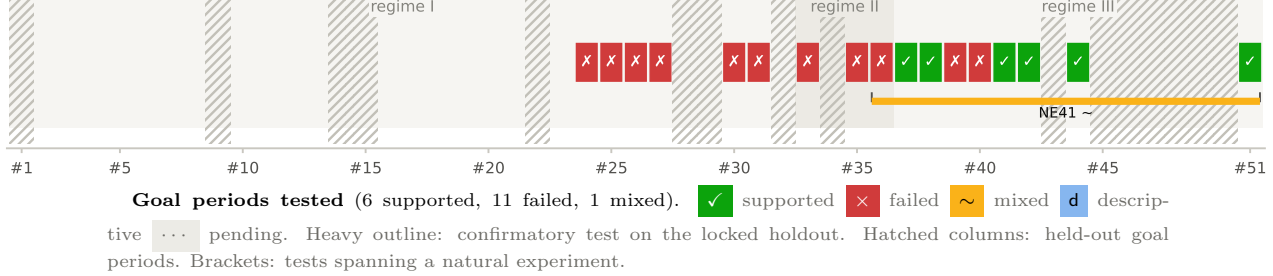
H08: Context is the coupling

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2.5/5 Usefulness 2.5/5

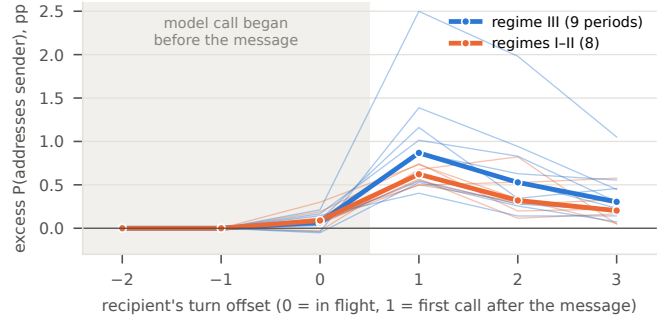
Bottom line. Agents respond only from the first model call after a message arrives, and erasing context cuts the coupling, but the nudge kernel is more than the read-out delay: “context is the coupling” fixes where responses land, not their full timing.

level: hypothesis · pipeline checks 71% · updated 2026-10-04



WHAT HAPPENED

Do agents couple only through what enters their model calls? We tested this in 17 non-holdout goal periods. (1) After a room message, the recipient’s chance of addressing the sender jumps at its first call that started after the message, with no jump at the call already running. This held in 10/11 regime-II/III periods (16/17 post hoc, for recipients not mid-exchange); regime I fails. (2) HH92 fails for nudges: #51 targets read the nudge in 1.7 min (median) but start working at ≈ 5 min, and a constant delay fits as well. (3) Forced context erasure (NE41) cuts addressing of senders read before it by 18% (9/9 periods; predicted 30%). (4) The Claude Code agent’s fetches match a room rule (recall 0.97–1.00) until 2026-03-17, when its feed began replaying April 2025. Hold-out script written, not run.



Observables. Excess chance that a recipient’s turn addresses the sender, by turn offset. Offset 0 is the call already running when the message arrived; offset 1 is the first call that started after it. Recipients not mid-exchange (post hoc); thin lines are periods, bold lines regime medians. The response appears only from the first call that can see the message, in every regime.

PHYSICAL MODEL

Two-variable kinetic Ising model (02) with Hawkes kernels (09). Each agent has a visible spin n_i (active per minute) and a hidden field c_i : the room events unread since its last model call. Calls start on the scaffold’s schedule $s_{i,k}$ (after the previous action, or when a pause timer expires). At each call start the field is read and reset, and couplings J_{ij} act only then. W : read-out delay; $h(u)$: post-read-out headroom; o : turn offset; β_F : coupling change after erasure.

$$c_i(t) = \{e : \text{room}(e) = r_i, s_i^- < t_e \leq t\}$$

$$P(a_i | s_{i,k}) \propto \exp \left[a \left(h_i + \sum_{e \in c_i(s_{i,k})} J_{i,\text{src}(e)} \right) \right]$$

$$G(\tau) = A \sum_w p(w) \Theta(\tau - w) h(\tau - w)$$

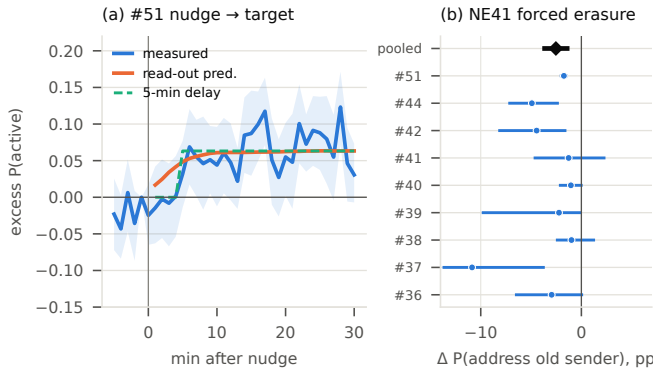
$$D = G_{\text{addr}}(o=1) - G_{\text{addr}}(o=0)$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An agent cannot react until its next model call starts, so the swarm’s coupling delay is its call cadence: tens of seconds when busy, minutes when paused. That follows from how the scaffold is built, so the news is modest; the value is quantitative and diagnostic. Compaction cuts about a fifth of the conversational coupling. A nudge takes about 5 min to show, so don’t expect instant effects. And the replay bug shows exposure can silently come loose from the room, so log what each agent actually fetched. HH92 failed; the gating result rests on mentions.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
C9 talk jumps at read-out ($\geq 70\%$ of periods)	6/17 (6/9 regime III, 0/8 partly regime I/II)	partly
C9 addressing jumps at read-out	11/17 (10/11 regime II/III); partly clean subset 16/17	
C9 other-room messages give no jump	6/8 two-room periods	yes
C8 nudge onset = read-out CDF	#51: $\Phi(1, 5)$ 0.03 measured vs 0.52 predicted	no
C8 context beats rivals in CV	beats immediate (0.77), no Hawkes (0.80), not a 5-min delay (0.41)	
C8 non-paused targets respond sooner	-0.02 [-0.09, 0.05]	no
NE41 erasure cuts coupling $\geq 30\%$	-18% $\pm$ 6%; negative in 9/9 periods	sign only
C1 room rule recall ≥ 0.9	0.97-1.00; 0.31 and 0.00 after the replay	until replay
C1 types differ, delay minutes	uniform coverage (WAITs too); refuted 14-23 s	
HH91: later half-day hotter (#51)	$\hat{n}$ 0.70 vs 0.50, Δ +0.20 [0.10, 0.29]	yes



Second figure. (a) #51 nudge kernel (95% band) with the zero-parameter read-out prediction and the best constant delay. The median read-out is 1.7 min, but the response starts at about 5 min. (b) NE41: change in the chance of addressing a sender read before a forced erasure, relative to one read after it (same turns, equal age), per period and pooled.

GOAL PERIODS IN DETAIL

G24 $\times$ failed
 G25 $\times$ failed
 G26 $\times$ failed
 G27 $\times$ failed
 G30 $\times$ failed
 G31 $\times$ failed
 G33 $\times$ failed
 G35 $\times$ failed
 G36 $\times$ failed
 G37 $\checkmark$ supported
 G38 $\checkmark$ supported
 G39 $\times$ failed
 G40 $\times$ failed
 G41 $\checkmark$ supported
 G42 $\checkmark$ supported
 G44 $\checkmark$ supported

+2 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Responses are @-mentions; the discontinuity removes common causes, not the proxy. The regime-I call rule is doubtful (addressing peaks at turns classed in flight), so its failures may be logging, not absent coupling. C8 rests on one powered period (#51, 303 nudges). Synthetic runs show the CV picks the right mechanism only about 2/3 of the time at this signal, and H04's matching leaves a pre-kick imbalance. Ground truth covers one agent with a different scaffold. The HH91 half-day effect may be within-day nonstationarity. No multiplicity correction; the C9 addressing and NE41 signs would survive one. Nothing confirmed on the holdout.

ROUND 2

Decompose the nudge lag into read-out plus a k-turn post-read-out lag, using talk and action turns. Swap mentions for a content-based reply measure. Fix the regime-I call rule. Test whether saving a sender to memory protects the thread across an erasure. Run the holdout script.

- **Essence.** Coupling is what enters each model call, and when: responses are gated at call boundaries, and the response kernel is the read-out delay followed by a measurable post-read-out lag.
- **H08-R1** Two-stage kernel: read-out delay convolved with a post-read-out lag counted in turns (talk and action turns aligned on the read-out turn), to decompose the nudge's 5-min onset and retest HH92 / H04-R1.
- **H08-R2** A non-mention response for the C9 discontinuity (content similarity to the sender's message against a message-specific null), so the gating result does not rest on names.
- **H08-R3** Regime-I call logging: identify which calls produce chat turns and redo the read-out rule, since addressing peaks at turns currently classed as in flight.
- **H08-R4** NE41 dose: whether writing a sender or thread to memory at the erasure protects the coupling (needs memory text: stored-line overlap with sender names).
- **H08-R5** Exposure audit: detect stale or replayed feeds (like the Claude Code agent's from 2026-03-17) wherever input logs exist, as a monitor for silent decoupling.